

Thamil Murugesan

Senior AI Engineer | AI & LLM Applications | Full Stack | TypeScript · AWS Bedrock · MCP

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SUMMARY

- 11 years building production web and mobile products in TypeScript, with a focus on shipping LLM features in regulated fintech.
- Shipped AI to production at OFX: AWS Bedrock-based agent that fixes accounting sync errors, an AI OCR platform that replaced third-party vendors, and Xero/QuickBooks MCP integrations for a company-wide AI support assistant.
- Hands-on with agentic patterns, including tool calling, MCP integration, and grounding LLMs in live data from internal APIs and Datadog.
- Builds AI features to production standard: event-driven AWS backends, retries and error handling, observability, analytics, and progressive rollout behind feature flags.
- Deep integration and identity experience: built Xero and QuickBooks two-way sync from scratch, implemented UK Open Banking (AISP/PISP), and led a company-wide Okta SSO migration across web and mobile.
- Owns delivery end-to-end, from UI to infrastructure; established a frontend framework adopted across product teams and mentored junior engineers.

SKILLS

AI & LLM: AWS Bedrock, Model Context Protocol (MCP), AI agents and tool calling, LLM document extraction/OCR, prompt engineering, RAG, LangGraph.js, Azure OpenAI

Frontend: React, Angular, Redux, TypeScript, Storybook

Mobile: React Native, Ionic, Capacitor

Backend: Node.js, Microservices, Serverless Architecture, DynamoDB, PostgreSQL, Redis

Cloud & Infrastructure: AWS (Lambda, Bedrock, API Gateway, EventBridge, CloudFormation, EC2, S3), Docker, Kubernetes, Pulumi

Integrations: Xero API, QuickBooks Online API, REST, webhooks

Security & Authentication: Okta, OAuth 2.0, SAML

DevOps & CI/CD: GitHub Actions, GitLab CI/CD, Jenkins, Netlify, LaunchDarkly

Testing, Analytics & Monitoring: Cypress, Playwright, BrowserStack, Datadog, CloudWatch, Mixpanel

CMS: Adobe Experience Manager (AEM)

EXPERIENCE

OFX, Sydney - Senior Software Engineer

Jun 2021 - Present

- Built Xero and QuickBooks Online MCP integrations for OFX's company-wide AI support assistant, a chat tool connected to Datadog and internal APIs. This lets teams investigate customer accounting-sync issues through natural-language queries.
- Built an AI accounting workflow agent on AWS Bedrock that diagnoses and resolves sync errors between OFX and Xero, instrumented with Mixpanel to track usage and resolution rates.
- Built an AI OCR platform that replaced third-party OCR providers; added merchant context to extraction, cutting failures by 30%.

- Built the QuickBooks Online integration from scratch and extended Xero, enabling two-way sync with the OFX app.
- Designed event-driven AWS Lambda services for accounting sync and Open Banking, with retries and error handling.
- Implemented UK Open Banking (AISP and PISP) and Accounts Payable features (bills, approval workflows, accounting logic).
- Shipped features safely through GitHub Actions CI/CD and LaunchDarkly progressive rollouts.
- Established a frontend framework and shared React component libraries (Storybook, internal npm) adopted across product teams.
- Led the migration from a legacy identity server to Okta SSO across web and React Native apps, including custom MFA flows.
- Built session management that let Okta coexist with the legacy auth server, plus Lambda migration services for a staged cutover with no disruption.
- Moved from frontend to full stack; presented at engineering guilds and took part in internal hackathons.

Stack: React, Node.js, React Native, AWS (Bedrock, Lambda, API Gateway), Okta, Pulumi, GitHub Actions, LaunchDarkly, Mixpanel, Cypress

Woolworths – Senior Software Engineer (TCS)

Nov 2019 - Jun 2021

- Worked on the migration of Woolworths Rewards, one of Australia's largest loyalty platforms, from AngularJS to Angular 9 – designing an incremental, non-breaking rollout strategy that preserved uptime and business continuity throughout.
- Architected a headless CMS using AEM Content Fragments with custom HTL components bridged to Angular via REST APIs, letting marketing publish content independently of development release cycles.
- Configured and tuned AEM Dispatcher as the caching and delivery layer – defining cache rules and TTLs for content-fragment and JSON endpoints, implementing flush and invalidation strategies to keep cached content current on publish, and hardening filter rules to restrict unsafe request paths to the publish tier.
- Built a centralised design system and reusable Angular component library with Storybook, establishing consistent UX standards and accelerating delivery across multiple application teams.
- Managed and streamlined Jenkins CI/CD pipelines, improving build stability and reducing manual deployment intervention.
- Mentored junior developers and contributed to frontend architecture decisions across technical chapters.

Stack: Angular 9, AngularJS, AEM, HTL, Jenkins, GitHub, Content Fragments, Storybook, SASS, REST APIs, Agile

TCS – Software Engineer

Aug 2015 - Nov 2019

Delivered enterprise web and mobile applications across financial services, workplace collaboration and education for clients in Australia and globally.

Citibank (CitiPlanner): Developed and maintained the front end of Citibank's wealth management platform, supporting portfolio management, holdings views and investment tracking. Built responsive, cross-browser interfaces with JavaScript, jQuery, Bootstrap and CSS, working to the bank's accessibility and browser-support standards. Resolved production defects and change requests through ServiceNow in a UNIX-based release environment, operating within banking change-control and security processes.

Fresco Docs: led front-end delivery of an enterprise document storage and sharing platform, building the React web client and React Native mobile app, implementing SAML-based authentication, and integrating against Node.js APIs owned by the backend team using Swagger.

Fresco Talk: Built a Slack-style enterprise messaging platform using React, Redux and Sagas, creating reusable components and optimising UI rendering performance.

Fresco Learn: Led a team of three delivering gamified web and mobile learning applications with a course-management CMS, working directly with business stakeholders to define requirements and running Agile delivery for the team.

GOIT Challenge: Built and shipped React Native STEM learning and quiz applications for Australian school students, released to both the App Store and Play Store, with Firebase-backed APIs, automated release pipelines via Fastlane and cross-device validation using BrowserStack.

Stack: React, Redux, Angular, AngularJS, Ionic, React Native, Node.js, Firebase, SAML, Webpack, Jenkins, AWS S3

EDUCATION

Bachelor of Technology (Information Technology)— Pondicherry University, India

Aug 2011 - May 2015

Score: 8.6/10

AWARDS AND RECOGNITION

- Two international IEEE research papers published on cloud security
- Best team award for Hackathon organizing - OFX
- Multiple TCS Awards for Excellence in Delivery